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Nguyen et al.

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(54) **STRAWBERRY PLANT NAMED 'DRISSTRAWONEHUNDREDTWO'**

(50) Latin Name: ***Fragaria* x *ananassa***
Varietal Denomination: **DrisStrawOneHundredTwo**

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A01H 6/74 (2018.01)
A01H 5/08 (2018.01)

(52) **U.S. Cl.**
USPC **Plt./208**

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USPC Plt./208
See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct variety of strawberry plant named 'DrisStrawOneHundredTwo', particularly selected for its yield potential, plant habit, and timing of fruit production, is disclosed.

6 Drawing Sheets

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STRAWBERRY PLANT NAMED 'DRISSTRAWONEHUNDREDTWO'

Latin name:
Botanical classification: *Fragaria* x *ananassa*.
Varietal denomination: The varietal denomination of the claimed variety of strawberry plant is 'DrisStrawOneHundredTwo'.

BACKGROUND OF THE INVENTION

Cultivated strawberry is a hybrid species of the genus *Fragaria* that is grown worldwide for its fruit. Modern strawberry was first bred in Brittany, France, in the 18th century by crossing *Fragaria virginiana* with *Fragaria chiloensis*. Strawberry fruit is an aggregate accessory fruit, with the fleshy part of the fruit being derived from the receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape, flavor, season of ripening, degree of fertility, and susceptibility to disease. Certain varieties vary in foliage, and some vary in the relative development of their reproductive organs. Typically, strawberry flowers appear hermaphroditic in structure, but function as either male or female. Generally, commercial production of strawberry plants involves propagation from runners and distribution as either plugs or bare root plants. Cultivation is either perennial or annual plasticulture. During the off season, strawberries can also be produced in greenhouses.

Strawberry fruit is widely appreciated for its characteristic bright red color, aroma, juicy texture, and sweetness. Strawberry fruit is a popular fruit that is generally consumed either fresh or in prepared foods, such as preserves and baked goods.

Strawberry is an important and valuable fruit crop. Accordingly, there is a need for new varieties of strawberry plants. In particular, there is a need for improved varieties of strawberry plant that are stable, high yielding, and agronomically sound.

SUMMARY OF THE INVENTION

In order to meet these needs, the present invention is directed to an improved variety of strawberry plant. In particular, the invention relates to a new and distinct variety of strawberry plant (*Fragaria* x *ananassa*), which has been denominated as 'DrisStrawOneHundredTwo'.

Strawberry plant variety 'DrisStrawOneHundredTwo' originated from a controlled cross between the proprietary strawberry female parent '100AB158' (unpatented) and the proprietary strawberry male parent '89AC29' (unpatented). Progeny plants from this cross, including 'DrisStrawOneHundredTwo', were asexually propagated via stolons in Shasta County, Calif. in April 2016. Strawberry plant variety 'DrisStrawOneHundredTwo' was later specifically identified and selected in Ventura County, Calif. in October 2016.

'DrisStrawOneHundredTwo' was subsequently asexually propagated via stolons, and has undergone testing at test plots in Ventura County, Calif. for six years (2016 to 2022). The present variety has been found to be stable and reproduce true to type through successive asexual propagations via stolons and tissue culture.

'DrisStrawOneHundredTwo' was particularly selected for its yield potential, plant habit, and timing of fruit production.

DESCRIPTION OF THE DRAWINGS

This new strawberry plant is illustrated by the accompanying photographs. The colors shown are as true as can be reasonably obtained by conventional photographic procedures. Unless otherwise indicated, the photographs are of plants that are six months old.

FIG. **1** illustrates whole fruit of variety 'DrisStrawOneHundredTwo'.

FIG. **2** illustrates longitudinal sections of fruit of variety 'DrisStrawOneHundredTwo'.

FIG. **3** illustrates the upper surface (top row) and lower surface (bottom row) of flowers of variety 'DrisStrawOneHundredTwo'.

FIG. **4** illustrates the lower surface (top row) and upper surface (bottom row) of leaves of variety 'DrisStrawOneHundredTwo'.

FIG. **5** illustrates an aerial view of a whole plant of variety 'DrisStrawOneHundredTwo'.

FIG. **6** illustrates a side view of whole plants of variety 'DrisStrawOneHundredTwo'.

DETAILED BOTANICAL DESCRIPTION

The following detailed descriptions set forth the distinctive characteristics of 'DrisStrawOneHundredTwo'. The data which define these characteristics is based on observations taken in Ventura County, Calif. from 2016 to 2022. This description is in accordance with UPOV terminology. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic, and cultural conditions. 'DrisStrawOneHundredTwo' has not been observed under all possible environmental conditions. The botanical description of 'DrisStrawOneHundredTwo' was taken from plants that were six months old. The indicated values represent averages calculated from measurements of several plants. Color references are primarily to The R.H.S. Colour Chart of The Royal Horticultural Society of London (R.H.S.) (2015 edition). Descriptive terminology follows the *Plant Identification Terminology, An Illustrated Glossary*, 2nd edition by James G. Harris and Melinda Woolf Harris, unless where otherwise defined.

Classification:
- *Species.*—*Fragaria* x *ananassa*.
- *Common name.*—Strawberry.
- *Denomination.*—'DrisStrawOneHundredTwo'.

Parentage:
- *Female parent.*—Proprietary strawberry plant '100AB158' (unpatented).
- *Male parent.*—Proprietary strawberry plant '89AC29' (unpatented).

Plant:
- *Height.*—23 cm.
- *Diameter.*—30.8 cm.
- *Height/width ratio.*—0.75.
- *Number of crowns per plant.*—6.
- *Growth habit.*—Semi-upright.
- *Density of foliage.*—Medium.
- *Vigor.*—Medium.

Stolon:
- *Diameter (at bract).*—3.2 mm.
- *Overall color.*—RHS 144A (Strong yellow green).
- *Anthocyanin coloration.*—Strong.
- *Anthocyanin color.*—RHS 59C (Moderate purplish red).
- *Density of pubescence.*—Medium.

Fruiting truss:

Length (*from crown to base of terminal flower or fruit*).—18.7 cm.

Diameter (*at base of truss*).—0.96 cm.

Number of berries per truss.—5.

Attitude at first picking.—Semi-erect.

Color (*at base of truss*).—RHS 144C (Strong yellow green).

Leaf:

Number of leaflets.—Three only.

Color of leaf upper surface.—RHS NN137B (Greyish olive green).

Color of leaf lower surface.—RHS 138C (Moderate yellow green).

Blistering.—Medium.

Glossiness.—Medium.

Variegation.—Present.

Terminal leaflet.—Length: 7.4 cm. Width: 6.8 cm. Length/width ratio: 1.09. Number of teeth per terminal leaflet: 21. Overall shape: Orbicular. Shape of base: Obtuse. Shape of apex: Rounded. Margin: Crenate. Margin profile: Revolute. Shape in cross section: Concave.

Petiole.—Length: 20.8 cm. Diameter: 4.2 mm. Overall color: RHS 143C (Strong yellow green). Pubescence: Dense. Attitude of hairs: Slightly outwards. Bract frequency (number present on each petiole): 0.

Petiolule.—Length: 7 mm. Diameter: 1.6 mm. Color: RHS 143C (Strong yellow green).

Stipule.—Length: 37.2 mm. Width: 8.9 mm. Stipule color: RHS NN137B (Greyish olive green). Anthocyanin coloration: Absent. Pubescence: Absent or very sparse.

Inflorescence:

Number of flowers per plant.—20.

Position of inflorescence in relation to foliage.—Same level.

Flowering interval.—November to June.

Pedicel.—Attitude of hairs: Slightly outwards.

Flower.—Flower diameter (petal tip to petal tip on non-flattened flower): 20.4 mm. Arrangement of petals: Touching. Size of calyx in relation to corolla: Larger. Receptacle color: RHS 145B (Light yellow green). Stamen: Present. Anther color: RHS 153B (Strong greenish yellow).

Petal.—Length: 9.0 mm. Width: 9.2 mm. Length/width ratio: 0.98. Number of petals per flower: 6. Color of upper and lower surface: RHS NN155A (Yellowish white). Overall shape: Orbicular. Shape of apex: Rounded. Margin: Entire. Shape of base: Convex.

Calyx.—Diameter (sepal tip to sepal tip, measured on back of flower): 30.8 mm.

Sepal.—Length: 11.6 mm. Width: 5.9 mm. Number of sepals per flower: 11. Overall shape: Elliptical. Margin: Entire.

Fruit:

Fruit size.—Length: 43.3 mm. Width: 41.8 mm. Length/width ratio: 1.04.

Fruit hollow.—Length: 18.8 mm. Width: 10 mm. Length/width ratio: 1.87.

Shape.—Conical.

Difference in shape of terminal and other fruits.—None or very slight.

Fruit color.—RHS N34A (Moderate red).

Evenness of color.—Even or very slightly uneven.

Glossiness.—Medium.

Evenness of surface.—Even or very slightly uneven.

Width of band without achenes.—Narrow.

Position of achenes.—Below surface.

Position of calyx attachment.—Level with fruit.

Attitude of sepals.—Upwards.

Diameter of calyx in relation to diameter of fruit.—Slightly smaller.

Adherence of calyx.—Strong.

Firmness.—Medium firm.

Color of flesh (*excluding core*).—RHS 41B (Strong red).

Evenness of flesh color.—Even.

Distribution of flesh color.—Marginal and central.

Color of core.—RHS 41C (Moderate reddish orange).

Sweetness/soluble solids (in ° Brix).—7.6.

Titratable acidity (*as % citric acid*).—0.54%.

Individual fruit weight.—31.0 g/fruit.

Achenes.—Number of achenes per fruit.—331. Weight: 0.00054 g/achene. Color of upper (sunward) side: RHS 53A (Deep red). Color of lower (shaded) side: RHS 152D (Dark greenish yellow).

Fruiting.—Harvest interval: December to May. Type of bearing: Not remontant. Productivity: 20,788 kg to 49,927 kg of fruit per hectare per season from six-month-old plants when grown in Ventura County, Calif.

Resistance to abiotic stress, pests, and diseases:

Two-spotted spider mite (*Tetranychus urticae*).—Moderately resistant.

Botrytis fruit rot (*Botrytis cinerea*).—Moderately resistant.

Powdery mildew (*Podosphaera macularis*).—Moderately resistant.

Anthracnose crown rot (*Colletotrichum acutatum*).—Moderately resistant.

COMPARISON WITH PARENTAL AND REFERENCE VARIETIES

'DrisStrawOneHundredTwo' differs from the female parent proprietary strawberry plant '100AB158' (unpatented) in that 'DrisStrawOneHundredTwo' has a more compact plant habit, higher yield potential, and a higher number of trusses as compared to '100AB158'.

'DrisStrawOneHundredTwo' differs from the male parent proprietary strawberry plant '89AC29' (unpatented) in that 'DrisStrawOneHundredTwo' has a more compact plant habit, higher yield potential, and a higher number of trusses as compared to '89AC29'.

'DrisStrawOneHundredTwo' differs from the reference variety 'DrisStrawThirtySix' (U.S. Plant Pat. No. 25,698) in that 'DrisStrawOneHundredTwo' has strong stolon anthocyanin coloration, dense petiole pubescence, small flower diameter, and the difference in shape of terminal and other fruits is none or very slight, whereas 'DrisStrawThirtySix' has medium stolon anthocyanin coloration, absent or very sparse petiole pubescence, large flower diameter, and the difference in shape of terminal and other fruits is moderate.

'DrisStrawOneHundredTwo' differs from the reference variety 'DrisStrawTwentySeven' (U.S. Plant Pat. No. 23,400) in that 'DrisStrawOneHundredTwo' has absent or very weak stipule anthocyanin coloration, early time of flowering, medium fruit size, and the fruit glossiness is medium, whereas 'DrisStrawTwentySeven' has weak stipule anthocyanin coloration, very early time of flowering, very large fruit size, and the fruit glossiness is strong.

We claim:

1. A new and distinct variety of strawberry plant named 'DrisStrawOneHundredTwo' as shown and described herein.

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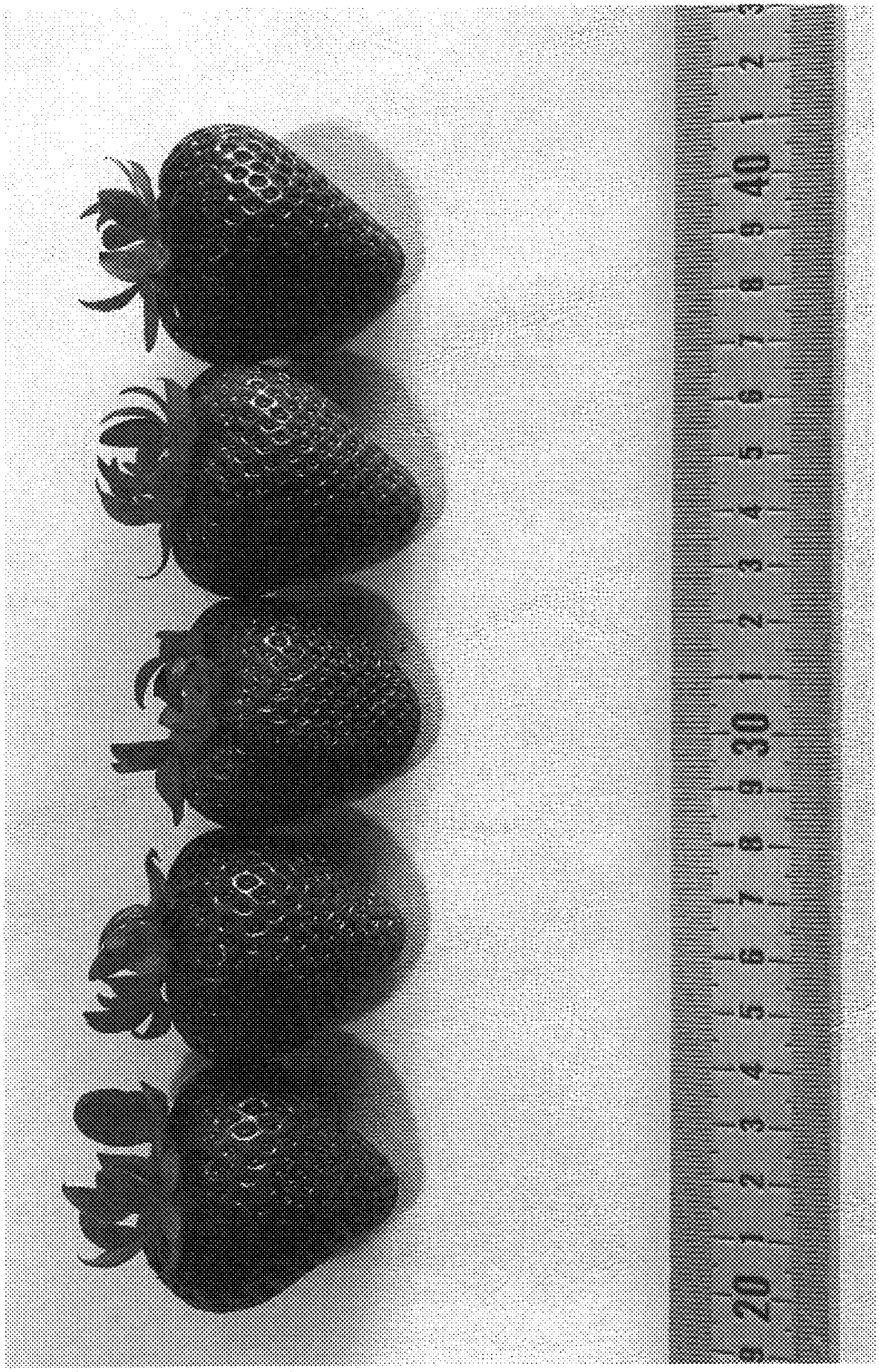

FIG. 1

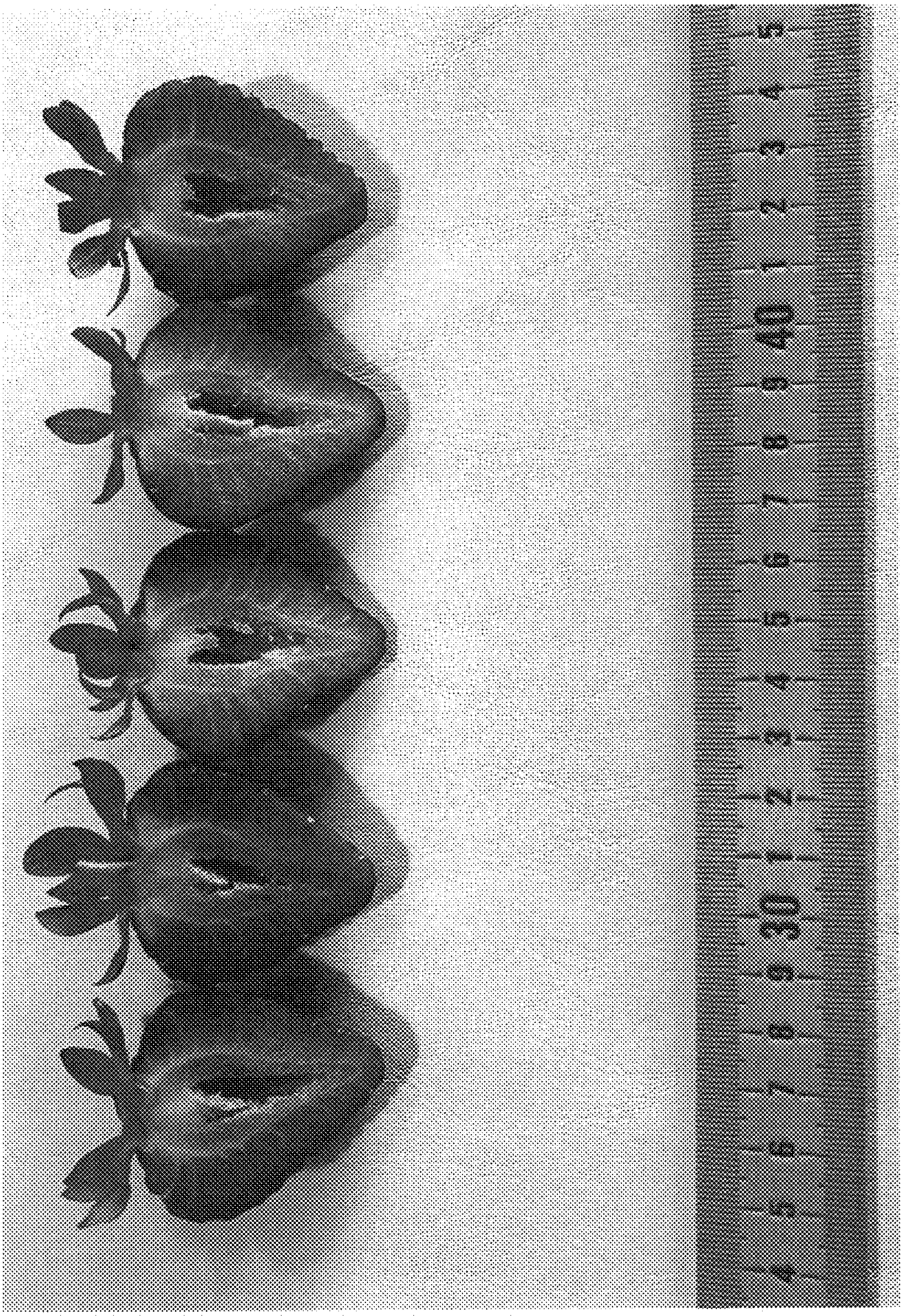

FIG. 2

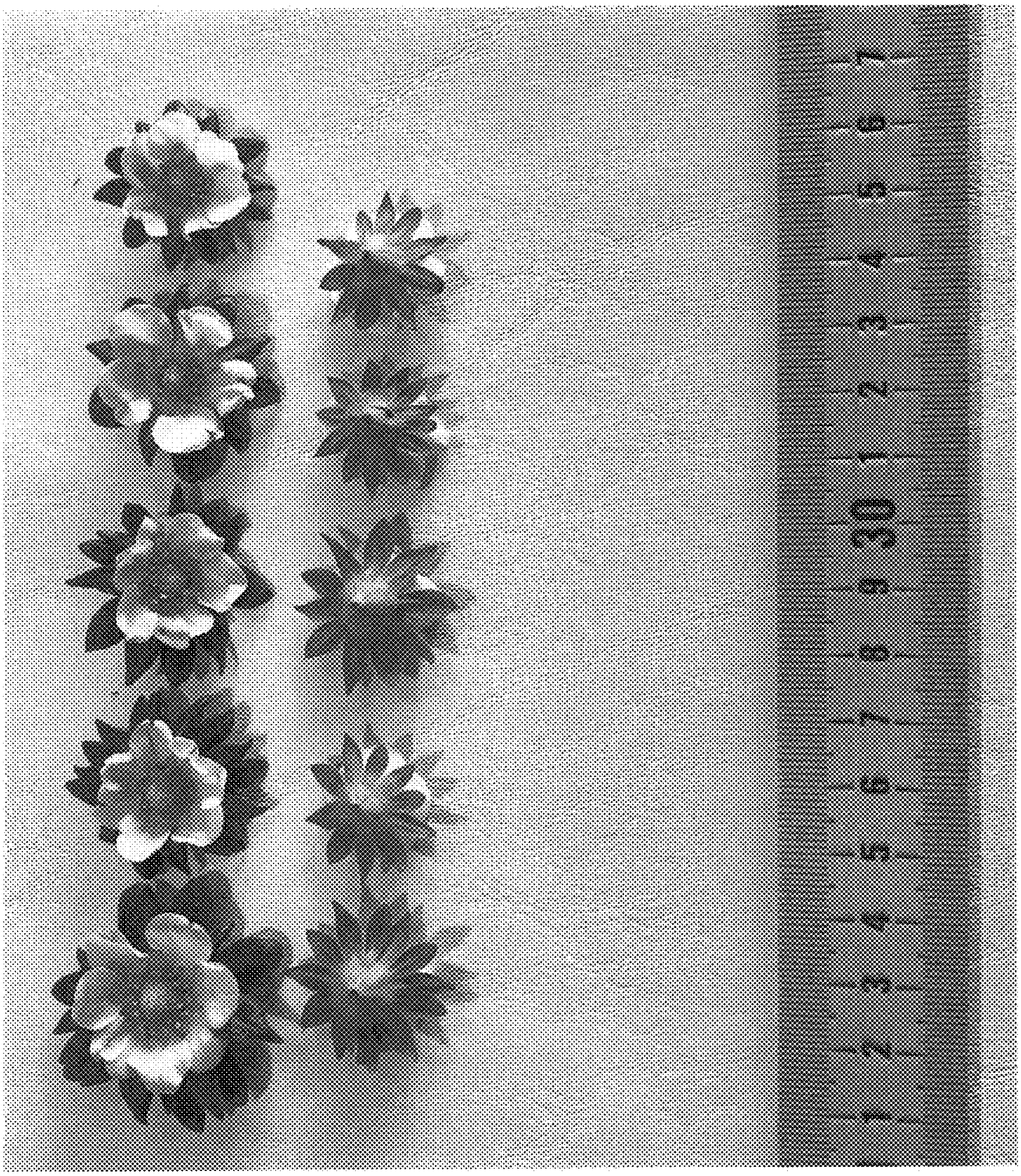

FIG. 3

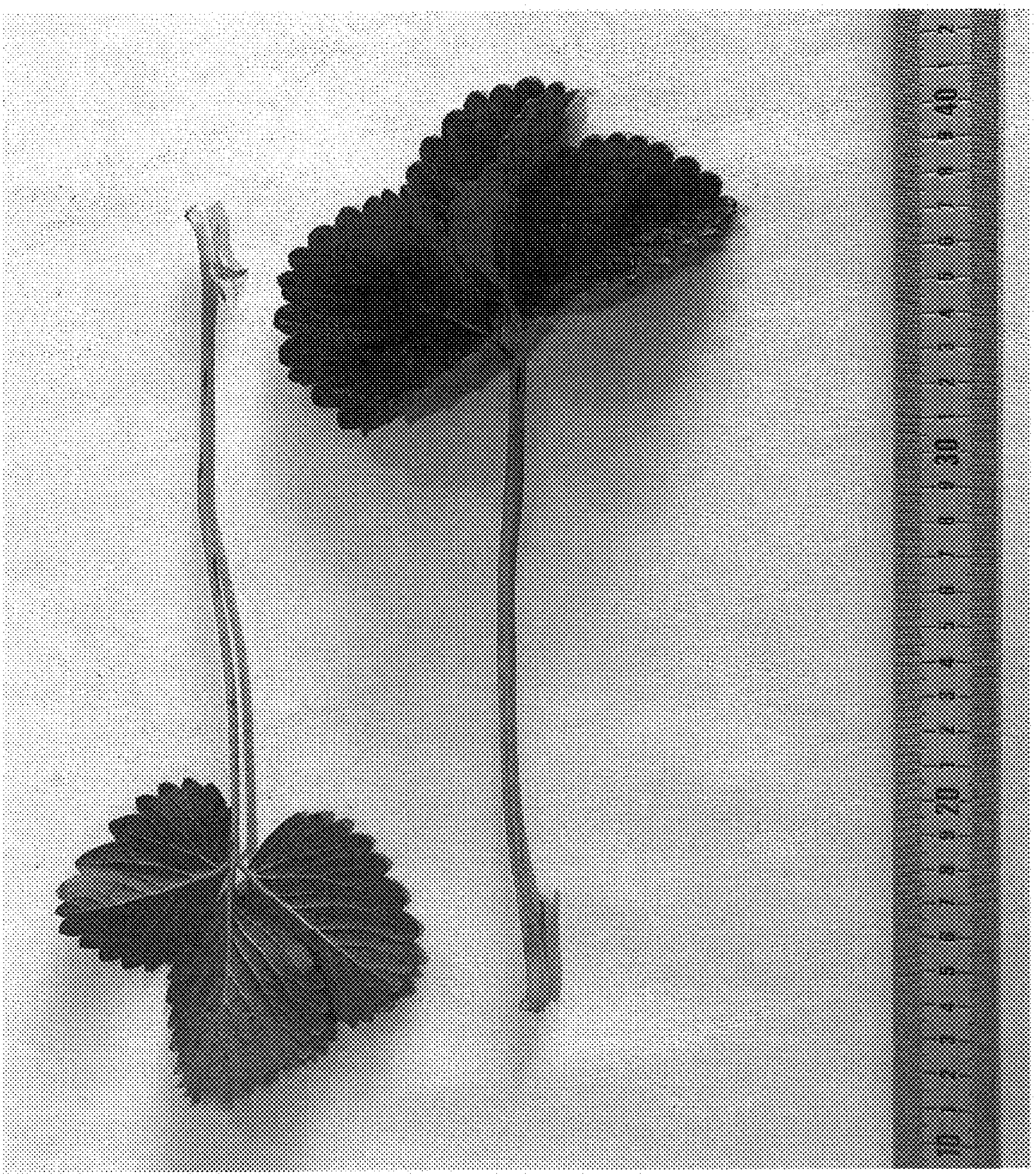

FIG. 4

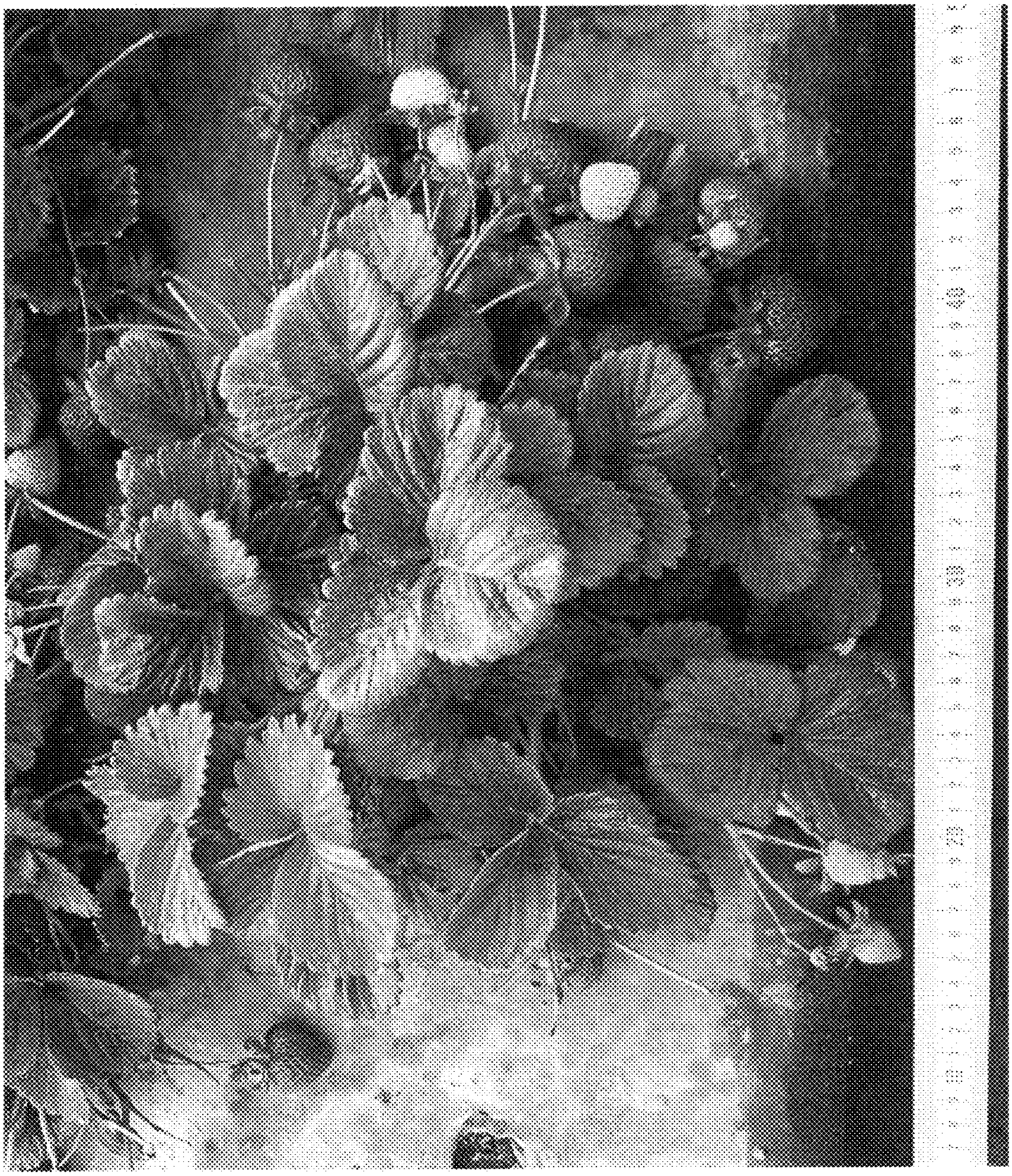

FIG. 5

FIG. 6